Project Proposal & Implementation Blueprint

Synthetic Identity Fraud Detection in Digital Lending Onboarding

1. Problem Statement Overview

Digital lending platforms enable rapid loan approvals through automated customer onboarding. However, this introduces severe exposure to **Synthetic Identity Fraud (SIF)**—where fraudsters construct composite identities combining real PII fragments with fabricated contact details.

Why Static KYC Fails: Individual PII fields pass isolated OCR/database checks. Furthermore, absence of immediate victims allows fraudsters to cultivate credit scores before maxing out credit limits.

ML Solution: We implement a 20-signal multimodal ML framework unifying KYC matching, identity age, behavioral biometrics (keystroke cadence, paste ratio), and device velocity into a real-time risk engine.

2. Team Roles & Task Allocation

Role / Area	Key Responsibilities	Deliverables Created
1. Lead AI/ML Engineer	Model selection, 5-fold Stratified CV, hyperparameter tuning, metric saving.	train_model.py, fraud_model.joblib, metrics.json
2. Data & Feature Engineer	20-signal synthetic dataset design (10,000 apps), noise modeling, distributions.	generate_data.py, synthetic_kyc_behavioral.csv
3. Full-Stack / UI Engineer	Streamlit research portal, live risk inference engine, batch telemetry inspector.	dashboard.py (Streamlit Web Portal)
4. Research & Doc Lead	Literature survey, formal IEEE paper drafting, 12-slide presentation deck, charts.	research_paper.pdf, presentation.pptx, roc_curves.png

3. Implementation Roadmap

Phase & Timeline	Objectives & Activities	Phase Output
Phase 1 (Week 1)	Problem definition, literature survey, threat taxonomy formalization.	Problem Statement & Threat Model
Phase 2 (Week 1–2)	20-signal multimodal feature engineering, dataset generation (10,000 rows).	synthetic_kyc_behavioral.csv
Phase 3 (Week 2–3)	5-model benchmark suite (LR, DT, RF, HistGBDT, MLP) & 5-fold CV evaluation.	train_model.py, research_metrics.json
Phase 4 (Week 3–4)	Interactive Streamlit web portal development & real-time risk engine.	dashboard.py (Active Port 8501)
Phase 5 (Week 4)	Formal IEEE research paper compilation, slide deck, and final defense.	research_paper.pdf, presentation.pptx, ZIP archive

4. Benchmark Metrics & Repository

GitHub Repository: https://github.com/sgsatpute/bfsi
5-Fold Cross-Validation Metrics: ROC-AUC: **0.9139** | Fraud Precision: **0.8600** | Fraud Recall: **0.9110** | F1-Score: **0.8849**
Deliverables: Source Code, 10k Dataset, IEEE Research Paper (PDF/DOCX), 12-Slide PPTX Presentation, Streamlit Web Portal, Submission ZIP Package.